Introduction to Biostatistics: Part 2, Descriptive Statistics

Descriptive statistics include measures of central tendency and variability. Measures of central tendency include mean, median, and mode. The mean is the arithmetic average of data from interval or ratio scales. The median reflects the 50th percentile score. The mode is the most frequently occurring value of a data distribution. Measures of variability include range, interquartile range, standard deviation, and standard error of the mean. The range describes the spread between the extreme values of data. Interquartile range is data included between the 25th and 75th percentile of a distribution. Standard deviation describes variability of data about the sample mean, while standard error of the mean helps describe the distribution of several sample means about a true population mean. Finally, confidence intervals, which are derived from the standard error of the mean, define an interval likely to include a true population value, based on sample statistical values and probability characteristics of data distributions. [Gaddis GM, Gaddis ML: Introduction to biostatistics: Part 2, descriptive statistics. Ann Emerg Med March 1990;19:309-315.]

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INTRODUCTION

Statistical analysis is the process by which numerical data obtained by scientific inquiry are transformed into a useable form for scientific interpretation. This involves manipulation of data for describing characteristics studied (descriptive statistics) and transformation of the data to help infer conclusions from the data (inferential statistics).

This second of a six-part series on biostatistics focuses on descriptive statistics. A thorough understanding of this topic is needed before advancing to discussions about inferential statistics. Familiarity with the concepts regarding types of data and data distributions, as presented in part 1,[1] is required for understanding the concepts presented herein. Numerical examples are provided to facilitate understanding. Finally, there exist many common, yet inappropriate uses of statistics, which will be discussed in this article.

MEASURES OF CENTRAL TENDENCY

Mean

The mean is the arithmetic average of data and is expressed by the equation:

$$\bar{X} = \Sigma X_i/n$$

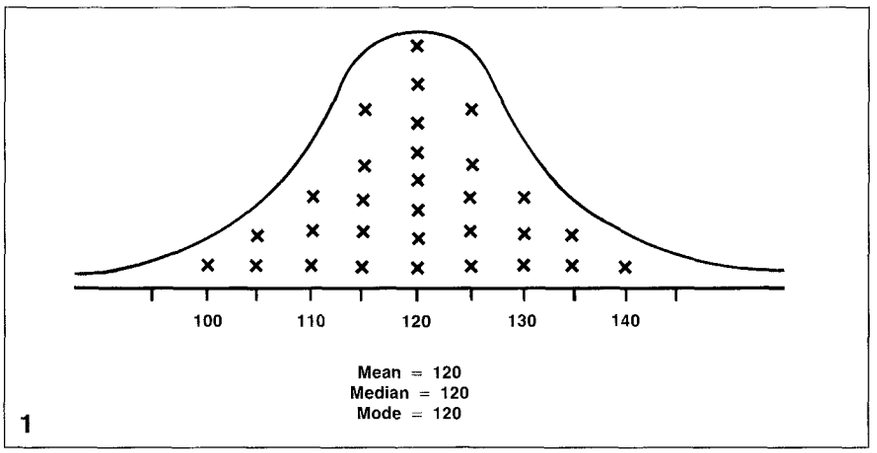
where $\bar{X}$ equals the mean, X_i equals each individual data point, and n is the number of data points in the sample. The mean can be calculated for interval and ratio scale data. However, mean values for ordinal scale data are generally misleading or invalid due to the lack of a consistent level of magnitude between numeric units of the scale.[1] Therefore, the mean is useful for data such as heart rate and blood pressure, but is misleading for arbitrarily constructed data scales such as the Apgar Scale, Glasgow Coma Score, and Trauma Score.[2,3]

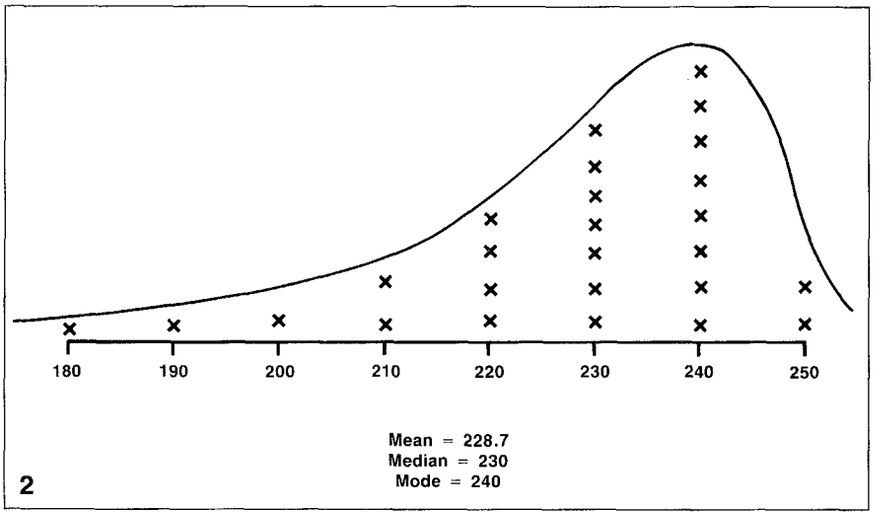
The mean is affected by outliers, which are extreme values of a data distribution.[2] This is not true of other measures of central tendency.

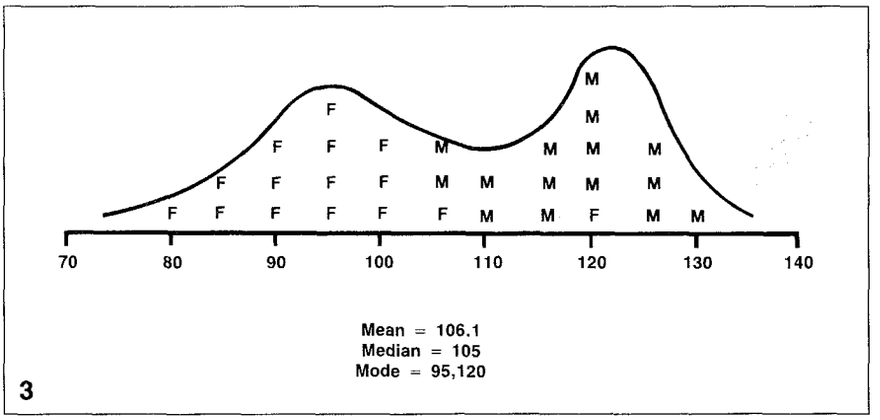
FIGURE 1. *Systolic blood pressure of 30 men aged 31 to 40 years. The mean, median, and mode all equal 120.*

FIGURE 2. *Systolic blood pressure in persons with renovascular hypertension. Mean, 228.7; median, 230; mode, 240.*

FIGURE 3. *Systolic blood pressure in young men and pregnant women. M denotes male subjects, F denotes female subjects. Mean, 106.1; median, 105; mode, 95, 120.*







Median

The median is the "mid-most" value of a data distribution. It is the value above which or below which half of the data points lie.[2,4,5] Alternatively, the median is the 50th percentile value of a distribution.

The median is unaffected by outliers and may be more useful than the mean to describe data when outliers exist[2] or when continuous data are not normally distributed.[4] The median is useful for describing ordinal data[4] because the magnitude of difference between points of a data scale need not be consistent to determine the 50th percentile value.[2] The median is not useful to describe nominal data[2] because of the arbitrary selection of numbers used to generate this scale.

Mode

The mode is the most commonly obtained value or values on a data scale, or the highest point of a peak on a frequency distribution.[2] The mode is most useful when two clusters of data exist (bimodal distribution), such that a group mean is misleading or meaningless.[2] The mode is useful to describe nominal data, defining the most prevalent characteristic of a sample.

Numerical Examples

Three different distributions of data will be examined to determine how the type of data distribution obtained affects the previously defined measures of central tendency.

Figure 1 represents normally distributed data for systolic blood pressure of 30 men aged 31 to 40 years. For normally distributed data, the values of the mean, median, and mode are identical.

Figure 2 presents theoretical systolic blood pressure data of patients with untreated renovascular hypertension. The distribution is negatively skewed. In the absence of normality, the mean, median, and mode are not equal. Also, an outlier, such as a systolic blood pressure value of 150 mm Hg instead of 180 mm Hg, will alter the value of the mean, but not the median or mode.

Figure 3 presents systolic blood pressure for a sample that includes two groups, pregnant women in their second trimester and men. Again, the mean and median are unequal in this non-normally distributed data. Also, there exist two peaks of data cluster,

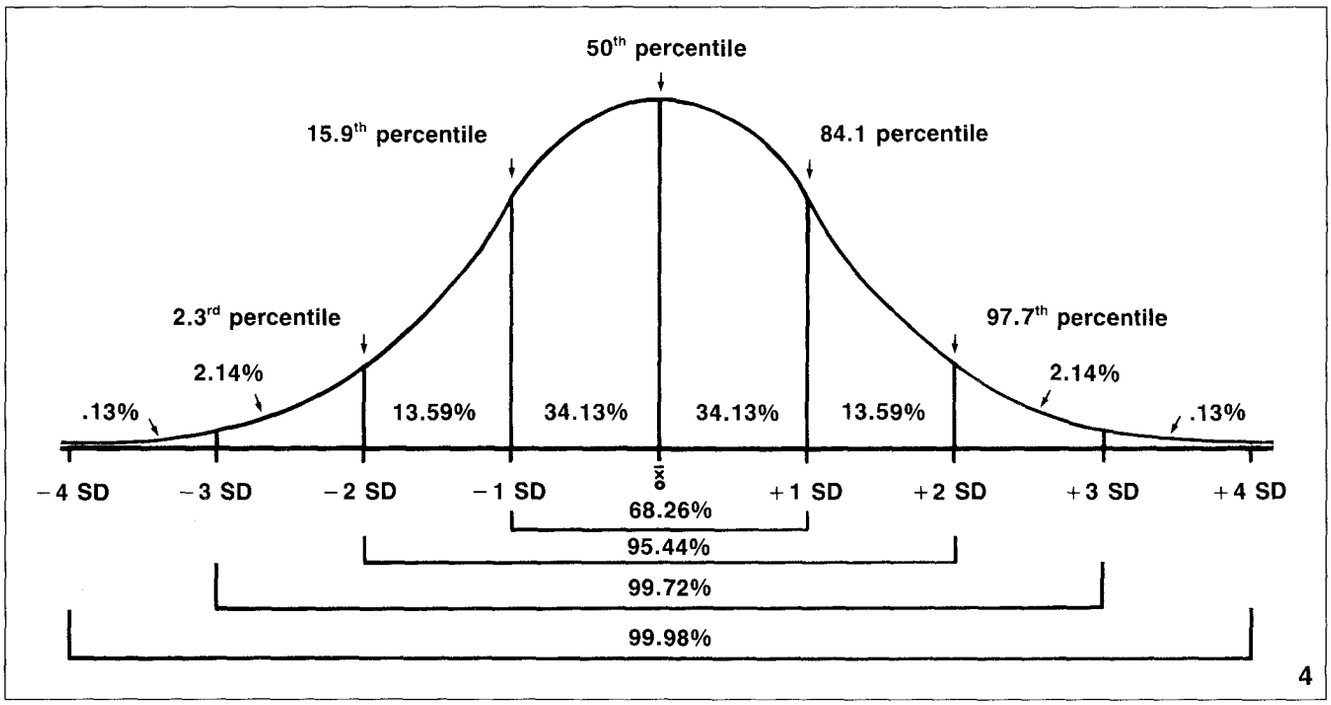


FIGURE 4. *SD and the normal distribution: 68.26% of all scores fall within ± 1 SD from the mean; 95.44% of all scores fall within ± 2 SD from the mean; 99.72% of all scores fall within ± 3 SD from the mean; 99.98% of all scores fall within ± 4 SD from the mean.*[5,6]

TABLE 1. *Applicability of measures of central tendency*

Characteristic	Mean	Median	Mode
Useful with interval, ratio data	Yes	Yes	Yes
Useful with ordinal data	No	Yes	Yes
Useful with nominal data	No	No	Yes
Affected by outliers	Yes	No	No

two modes. To ignore the bimodal aspect of this distribution would be to overlook its unique feature. Also, the presence of an outlier would alter the mean, but not the median or modes.

In Figures 2 and 3 the mean, median, and mode(s) are unequal because data are not normally distributed.[5] Thus, the measure of central tendency most useful to data analysis depends on the type of data, and what aspect of the data is to be conveyed. Fortunately, most physiologic data are normally or near normally distributed so that mean, median, and mode are equal. However, ordinal scale data have no consistent magnitude of difference between units of the data scale, and most ordinal data are not normally distributed.[3] Therefore, the mean is misleading as a measure of central tendency for ordinal scale data.[2,3]

No single measure of central tendency is best for all situations.[5] The applicability of measures of central tendency is summarized (Table 1).

MEASURES OF VARIABILITY

Measures of central tendency do not describe the variability, or spread, of data. Standardized estimates defining data variability are needed to help infer whether two groups studied differ significantly. In other words, measures of variability are used to help infer whether two or more groups studied are drawn from different populations. Several estimates of variability exist.

Range

The range is the interval between the lowest and highest values within a data group.[2] It is the simplest measure of variability to understand and identify. While simple, the range only considers the extreme values of a series of measures, and thus the presence of one outlier can markedly influence the range. The range is purely a descriptive tool and should not be used to infer whether groups differ statistically.

Interquartile Range

The interquartile range is a measure of variability directly related to the median. Recall that the median, a measure of central tendency applicable to ordinal and non-normally distributed data, is the middlemost value of a set of data. The median represents the 50th percentile. The interquartile range is that range described by the interval between the 25th and 75th percentile values.[6]

It has been suggested that the interquartile range be used for describing the variability of data that do not meet parametric analysis standards, such as ordinal scale data.[6] The interquartile range clearly defines where the middle 50% of measures occurs and indicates the spread of the data

TABLE 2. *Estimates of variability of systolic blood pressure data of men aged 31 to 40 years*

Subject	Systolic Blood Pressure	$(\bar{X} - X_i)$	$(\bar{X} - X_i)^2$
1	135	15	225
2	115	5	25
3	110	10	100
4	130	10	100
5	125	5	25
6	125	5	25
7	105	15	225
8	120	0	0
9	120	0	0
10	120	0	0
11	125	5	25
12	110	10	100
13	115	5	25
14	115	5	25
15	135	15	225
16	100	20	400
17	120	0	0
18	125	5	25
19	120	0	0
20	130	10	100
21	140	20	400
22	120	0	0
23	115	5	25
24	110	10	100
25	130	10	100
26	105	15	225
27	120	0	0
28	115	5	25
29	125	5	25
30	120	0	0

Mean = Σ X/n = 3,600/30 = 120
Median = 120
Mode = 120
Variance: $\Sigma (\bar{X} - X_i)^2/(n-1)$ = 2,550/29 = 87.931
SD = $\sqrt{87.931}$ = 9.377
SEM = $\frac{9.337}{\sqrt{n}}$ = 1.712

without using statistical techniques improperly.

Standard Deviation

The standard deviation (SD) is one of the most commonly encountered estimates of data variability and is integral to performance of inferential statistical techniques.[2] It provides an estimate of the degree of scatter of individual sample data points about the sample mean.

The usefulness of the SD lies in its properties as related to the Gaussian, or normal, distribution. The SD itself can be used to define an extreme score, such as the value that is exceeded by 5% or 95% of all scores from a sample of a population.[2] Figure 4 shows that 68.26% of data points of a normally distributed population fall within plus or minus one SD of the mean, and 95.44% of points fall within plus or minus two SD of the mean.[5]

The SD is calculated as the square root of another term called the variance. Because individual data points will fall both above and below the mean, the effect of direction of differ-

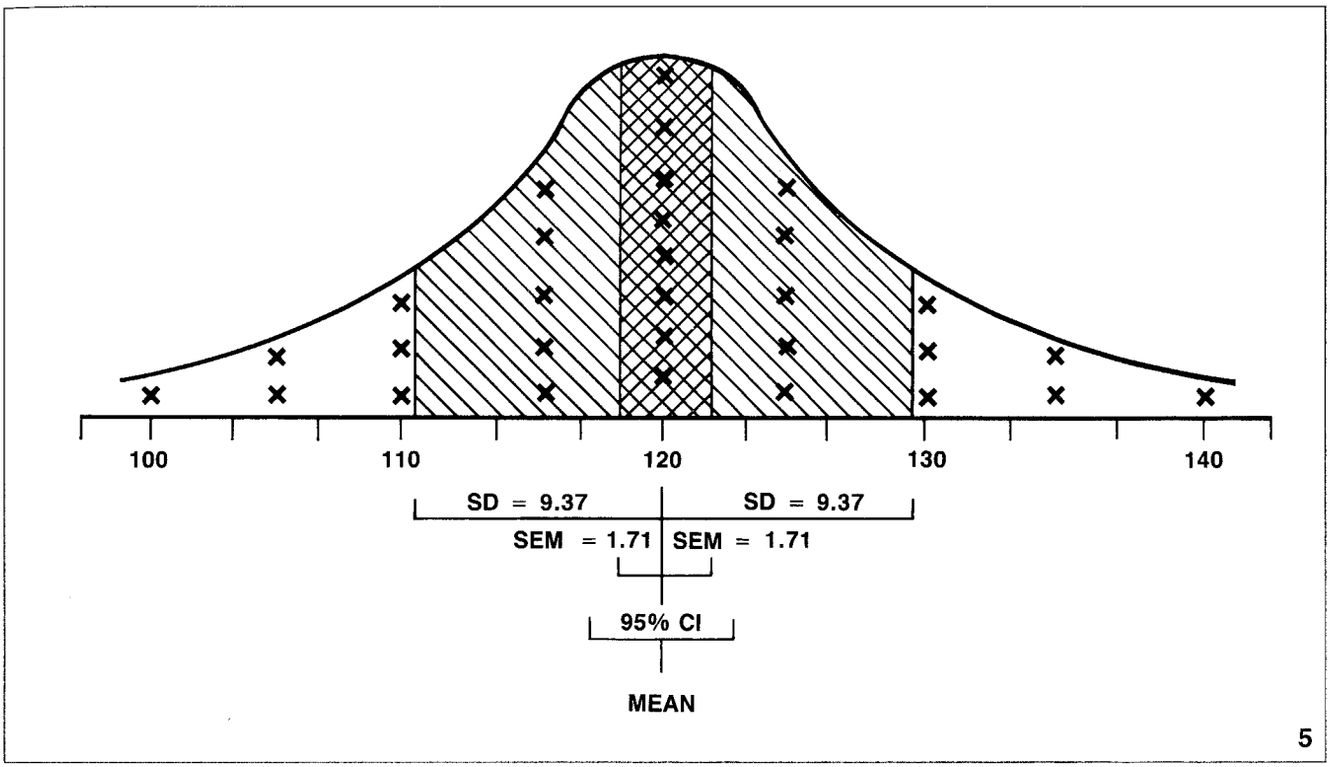


FIGURE 5. *Systolic blood pressure of men aged 31 to 40 years. SD, SE, and 95% CI are noted.*

TABLE 3. *Applicability of measures of variability*

Characteristic	Range	Interquartile Range	SD	SEM
Useful to describe interval or ratio data	Yes	Yes	Yes	Yes
Used to describe ordinal data	Yes	Yes	No	No
Descriptive of sample variability	Yes	Yes	Yes	No
Assists in statistical inference	No	No	Yes	Yes
Used to calculate confidence intervals	No	No	No*	Yes

*SEM = $SD/\sqrt{n}$, thus SD is involved indirectly in calculating a confidence interval.

ence will cause some deviations from the mean to be positive and some to be negative. To overcome this effect, deviations are squared to obtain a positive number. Individual squared deviations from the mean are then averaged to calculate the estimate of variability known as the variance. Numerically

$$\text{Variance} = \Sigma (\overline{X} - X_i)^2/(n-1)$$

where $\overline{X}$ equals the mean, X_i equals each individual data point, and n equals the total number of data points.[5-7] The variance represents the deviation from the mean, expressed as the square of the units used. For instance, in Figure 1, the variance of systolic blood pressure is expressed as mm Hg^2. However, these squared units are not meaningful. Therefore, the square root of the variance is then calculated, to bring the variability estimate back to the correct scale. This is the value known as the SD:

$$SD = \sqrt{\text{variance}}$$

The SD is meaningful only when applied to data that are normally or nearly normally distributed.[2,8,9] It is applicable to interval or ratio scale data.[2]

The SD is useful in application to statistical inference techniques. The calculation of SD from the normally distributed systolic blood pressure data of Figure 1 is shown (Table 2).

Standard Error of the Mean

The standard error of the mean (SEM) is a statistic derived from the SD, and is simply calculated as

$$SEM = SD/\sqrt{n}$$

It is obvious from the calculation that the SEM is always smaller than the SD and the greater the n, the smaller the SEM will be.

The SEM is an abstract concept. Imagine repeating an experiment numerous times. With each experiment, a different sample group would be drawn from the study population. Because each repetition of the experiment contains unique sample members, different mean values will be generated with each study. The collection of these mean values, as generated from repetitive sampling and experimentation, will reflect "scatter" about the true but unknown population mean. The SEM is simply a quantification of the variability of

these sample mean values. The SEM is properly used to estimate the precision or reliability of a sample, as it relates to the population from which the sample was drawn.[10,11] The SEM does *not* provide an estimate of the scatter of sample data about the sample mean[12] and should not be used as such.

The SEM is useful because it is used in the calculation of "confidence intervals," which contain an estimate of the true mean for an entire population from which the sample was drawn. Confidence intervals can be used for descriptive or inferential purposes.

A calculation of SEM for the normally distributed data presented in Figure 1 is shown (Table 2).

TABLE 4. *Effect of confidence level and sample size on confidence interval width*

Calculation of CIs for Data Presented in Table 2				
CI(%)	**SD**	**n**	**SEM**	**CI**
90	9.377	30	1.712	120 ± 2.82
95	9.377	30	1.712	120 ± 3.36
99	9.377	30	1.712	120 ± 3.83
Effect of Sample Size on CI for Data With A Mean of 120 and a SD of 9.377				
CI(%)	**SD**	**n**	**SEM**	**CI**
95	9.377	30	1.712	120 ± 3.36
95	9.377	100	0.938	120 ± 1.84
95	9.377	1000	0.297	120 ± 0.582

Standard Deviation Versus Standard Error of the Mean

Both SD and SEM are measures of variability. However, the two statistics are different and are frequently confused or misused.[12] The SD defines variability of sample data points about a sample mean. The SD is always greater than the SEM. The SEM is most commonly calculated to help derive confidence intervals.

Various authors have commented about the intellectual sleight of hand of incorrectly using SEM when only SD is appropriate to describe sample data variability.[6,12,13] Bunce et al[13] reviewed 608 articles in six journals in which mean ± SD or SEM were reported. In 50%, SEM values were reported when only the SD would have been appropriate. The authors concluded that "many workers may choose to report the SEM because it is simply smaller than the SD."[13] The inappropriate use of SEM to describe sample data variability may be presented by authors in an attempt to imply that a significant difference exists between groups, when in fact no difference exists. Elenbaas et al[12] were more blunt, concluding that authors who present data as mean ± SEM instead of mean ± SD may be trying to actively impair the reader's ability to accurately identify the variability in the study data.

Whether by error or by design, it is incorrect to underrepresent the variability of sample data as mean ± SEM. We suggest that readers multiply the SEM by $\sqrt{n}$ to obtain the SD when SEM is erroneously used to express sample variability. It is not an error to use SEM in speculating a range, or confidence interval, within which a true *population* mean is likely to fall. The SD and SEM of the data shown in Figure 1 are given (Figure 5). Table 3 summarizes the proper use of estimates of variability.

Confidence Intervals

When statistics derived from the sampling of a population are studied to infer values for population parameters, it would be useful to have confidence that the sample statistical value, such as a mean or SD, would be representative of the true population parameter. One cannot be certain that a sample statistical value is representative of the true population parameter, but one can calculate a range of values likely to be representative of the population parameter.[4,14] That range of values is called a confidence interval (CI). Calculation of a CI is a method of estimating the range of values likely to include the true value of a population parameter. Since one cannot study all members of a population, a representative sample of the population is studied, and from this one uses the mean and SEM to work backward to estimate a CI.

The width of the CI depends on the SEM and the degree of confidence we arbitrarily choose. For instance, a 95% CI, which is the degree of confidence most commonly expressed,[14] is a range of values broad enough that, if the entire population could be studied, 95% of the time the population mean would fall with the CI estimated from the sample of the population.[15] Also, the closer a point lies to the middle of the CI, the more likely it is representative of the population.[16]

Though by convention the 95% CI is most commonly reported, the 95% level is not rigidly required. Wider CIs, such as a 99% or 99.9% CI, are even more likely to include the true population parameter value and are commonly used for critical appraisal of data. They are also advocated for examinations of data during ongoing accumulation of subjects in a clinical trial.[15] Narrower CIs, such as the 90% CI, can be used when study authors find it acceptable that ten times out of 100, the true population parameter may not lie within the CI. However, the width of a CI depends not only on the variability of the data and the level of confidence selected, but also the sample size.

When one broadens a CI by moving a 95% to a 99% CI, accuracy is increased because the calculated CI becomes more likely to include a true population parameter. However, when the level of CI is held constant and sample size is increased, SEM is decreased and thus the CI is narrowed. This narrowing of the CI increases the precision of the CI. The effect of level of confidence selected and sample size on the width of a CI is shown (Table 4).

Calculation of the CI for estimation of true population mean values applies to continuous data from normal or near-normal distributions.[4] Also, a CI can be estimated for such other statistics as medians, regression slopes, relative risk data, re-

sponse rates, intergroup differences of response rates, X year survival rates, median survival duration, and hazard ratios.[14,16] In addition, CI may be used to visually compare data when two or more sample groups are studied but their members were not randomly selected or assigned between the groups.[15]

Pitfalls in the use of the CI exist. Confidence intervals convey the effects of sampling variation but do not control for such nonsampling errors in study design or execution as improper selection of subjects, poor experimental design, and the like.[4]

SUMMARY

This article has highlighted common proper and improper use of measures of central tendency and measures of variability. The relationship between the type of data scale and the correct use of mean, median, and mode have been presented. The variability estimates of range, interquartile range, SD, and SEM have been introduced, and their proper and improper use has been discussed. Finally, the concept and proper use of CIs have been outlined.

The next installment of this series, in the May issue, will cover hypothesis testing. Included will be types of experimental error, the terms alpha (α) and beta (β), statistical power, and sensitivity, specificity, and predictive value.

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